Part A – CV Building Task (5 marks)

Scenario

You are applying for a junior software developer position. Using LaTeX, build a **one-page, professional CV** for the candidate whose data is given below. Submit the *complete source code* (from `\documentclass` to `\end{document}`).

Candidate data

Raw data sheet

Name: Lionel Messi
Contact: lionel.messi@example.com | +880 1700 000000 | Dhaka, Bangladesh
Links: GitHub `https://github.com/messi` | LinkedIn `https://linkedin.com/in/messi`

Profile: Final-year Computer Science student with strong interest in algorithms, graph theory and software engineering. Seeking an entry-level developer role.

Education

- BSc in Computer Science and Engineering, Example University – 2021 to 2025 – CGPA 3.72/4.00
- Higher Secondary Certificate (Science), Example College – 2019 to 2021 – GPA 5.00/5.00

Experience

- Software Engineering Intern, TechNova Ltd. – Jun 2024 to Aug 2024, Dhaka
  - Built REST API endpoints in Python (Flask) used by a mobile app.
  - Reduced average query time by 30% by optimising SQL indexes.

Projects

- Campus Navigation App – shortest-path routing using Dijkstra's algorithm (C++).
- Library Management System – full-stack web application (Python, SQL).

Skills (level out of 5): Python 5, C++ 4, SQL 4, LaTeX 4, Git 3
Languages: Bangla (native), English (fluent)
Awards: Dean's List 2023 and 2024; 2nd place, University Programming Contest 2024

Tasks

A1. Page setup and title. Use the `article` class (11 pt, A4). Set margins to 2 cm with `geometry`, remove page numbers, and display the candidate's name as a large, centred, bold heading with the contact line and the two links below it. The links must be clickable (`hyperref`).

A2. Section structure. Create separate, clearly formatted sections for *Profile, Education, Experience, Projects, Skills, Languages* and *Awards*. Use `\section*` (or a custom styled heading) with a horizontal rule.

A3. Custom command. Define a command `\cventry{title}{date}{place}{details}` and use it for *every* education and experience item so that the title is bold and left-aligned, and the date is right-aligned on the same line (use `\hfill`).

A4. Lists. Present the intern responsibilities and the projects with `itemize` (use `enumitem` to reduce spacing).

A5. Skills table. Typeset the skills in a table using `booktabs` (`\toprule`, `\midrule`, `\bottomrule`) with columns *Skill* and *Level*. Draw the level as filled and empty circles (e.g. ●●●●○ for 4/5).

A6. Overall quality. The CV must compile without errors, fit on **one page**, and be visually consistent (fonts, spacing, alignment).

Part B – Graph Task: Nodes and Edges (5 marks)

Given graph

Consider the undirected, weighted graph $G = (V, E)$ with

$$V = \{A, B, C, D, E, F\}, \qquad E = \{AB, AC, BC, BD, CE, DE, DF, EF\}.$$

The edge weights are $w(AB) = 4,\ w(AC) = 2,\ w(BC) = 1,\ w(BD) = 5,\ w(CE) = 9,\ w(DE) = 2,\ w(DF) = 6,\ w(EF) = 3$.

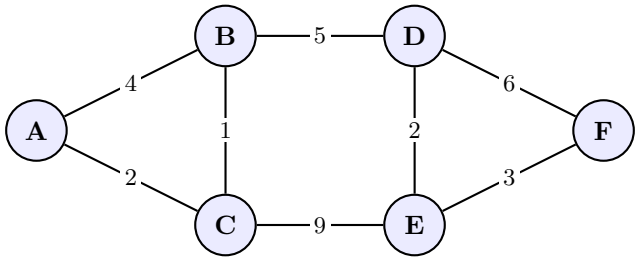


Figure 1: Weighted undirected graph G.

B1. Drawing with TikZ. Write the complete LaTeX code (the `tikzpicture` environment only) that reproduces Figure 1. Requirements:

(a) Define a node style `vertex` (circle, thick border, light fill).

(b) Place the six nodes at the coordinates $A(0, 2),\ B(2, 3),\ C(2, 1),\ D(4, 3),\ E(4, 1),\ F(6, 2)$.

(c) Draw all eight edges and write the weight on each edge (hint: `node[midway,fill=white]`).

(d) Highlight the edge AB in red and make it dashed.